Evaluating LLM-as-a-Judge for Hallucination Detection

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Research Qn 1

What is the hallucinated detection accuracy of LLM judges against human-verified labels, across the 3 different correctness conditions: correct, hallucinated and fabricated?

Research Qn 2

Does fluency or expressed confidence of an answer affect the likelihood that an LLM judge labels a hallucination as correct, even after controlling for factual correctness and question- level random effects?

Research Qn 3

How much of LLM judge error is attributable to answer style versus the inherent difficulty of detecting a given hallucination, and can that difficulty be estimated as a latent parameter using IRT?

Dataset – SQuAD 2.0

Features:

- 11.9K **samples**; each consists of a title, passage, question, and answer
- 35 unique **titles**
- 21 – 49 **context (passage)** per title
- 3 – 28 related **questions** per context (passage)
- 0 – 6 potential manually checked **answers** per question
- Avg. length of context passage: 127 words

```
{
    'id': '56de49a8cffd8e1900b4b7a8',
    'title': 'Normans',
    'context': "Bethencourt took the title of King of the Canary Islands,
as vassal to Henry III of Castile. In 1418, Jean's nephew Maciot de
Bethencourt sold the rights to the islands to Enrique Pérez de Guzmán, 2nd
Count de Niebla.",
    'question': 'Who bought the rights?',
    'answers': {
        'text': ['Enrique Pérez de Guzmán', 'Enrique Pérez de Guzmán',
'Enrique Pérez de Guzmán'],
        'answer_start': [172, 172, 172]
    }
}
```

Characteristics:

2 types of questions:

- **Answerable** – answer can be directly found in the passage, populated “answer” in dataset
- **Unanswerable** – answer is related to title, but cannot be found anywhere in the passage, NULL “answer” in dataset

2 types of answerable questions:

- **Same answer** – all provided answers are identidal
- **Different answer** – slight variation in provided answers

```
    'question': 'What is Norman art's most well known piece?',
    'answers': {
        'text': ['Bayeux Tapestry', 'the Bayeux Tapestry', 'the Bayeux
Tapestry'],
        'answer_start': [49, 45, 45]
    }
```

Inclusion criteria:

1. Context length: **<= 150 words**
2. For answerable qns, only those with **same answer**

Stratified sampling

- POOL A – 140 **answerable** qns
- POOL B – 140 **unanswerable** qns
- 35 titles X 4 context (passage) X 1 qn

Manual checking

- Verify each hallucination is actually incorrect, including those referencing the given context

Evaluate

- Determine whether each answer is **True / False**, based on given context
- Assign a score from **1 to 5** (completely unsupported by context to fully supported by context)
- False: 1 – 3, True: 4 – 5

01

02

gemini-2.5-flash

03

gemini-2.5-flash

04

Judge 1: gpt-4o-mini

Judge 2: claude-haiku-4-5-20251001

05

Generate wrong answers

- Create wrong answers strictly from the provided context (passage), **no outside knowledge** allowed
- Wrong answers may contradict the reference answer or fail to answer the question validly

Rewrite answers

- Rewrite each answer as a full-sentence response in different stylistic forms
- **Low fluency** – short, hesitant, and hedged wording that sounds less polished
- **High fluency** – longer, smoother, and more confident wording that sounds more assured
- **Attack** – persuasive wording that uses details from the passage to make wrong answers sound convincing

Pool A – Overview

Answer types:

- **Correct** – correct answer directly from source dataset
- **Incorrect, external** – wrong answer that contradicts the reference answer and is not supported anywhere in the context (passage)
- **Incorrect, internal** – wrong answer that contradicts the reference answer but is based on content found in the context (passage)


```
{
    'id': '56de3d594396321400ee26ca',
    'title': 'Normans',
    'context': 'Even before the Norman Conquest of England, the Normans had come into contact with Wales. Edward the Confessor had set up the aforementioned Ralph as earl of Hereford and charged him with defending the Marches and warring with the Welsh. In these original ventures, the Normans failed to make any headway into Wales.',
    'question': 'Where was Ralph earl of?',
    'correct': 'Hereford',
    'incorrect_ext': 'Shrewsbury',
    'incorrect_int': 'Marches'
}
```


Answer styles:

answer	style	ground_truth_ind	# Qns
Correct	low	1	140
	high	1	140
Incorrect, External	low	0	140
	high	0	140
	attack	0	140
Incorrect, Internal	low	0	140
	high	0	140
	attack	0	140

Wilcoxon Sign-ranked test

- Checks if rewriting answers in different styles changes their hedging, confidence, and length
- Hedging/confidence counts based on keywords specified in prompt templates

Stylistic feature	Statistically significant
Hedging	Low > High; Attack similar to High
Confidence	High > Low; Attack similar to Low
Word count	Attack ≥ High > Low

Pool A – Analysis (answerable)

01 What is the hallucinated detection accuracy of LLM judges against human-verified labels, across the 3 different correctness conditions: **correct**, **hallucinated** and fabricated?

Metrics

Metric	Answer Type	Judge 1 (GPT)	Judge 2 (Claude)
FRR	Correct	0.043	0.129
FAR	Incorrect (all)	0.110	0.073
FAR	Incorrect (External)	0.036	0.021
FAR	Incorrect (Internal)	0.183	0.124
Cohen's κ	Correct vs Incorrect (External)	0.920	0.864
Cohen's κ	Correct vs Incorrect (Internal)	0.745	0.741

Tests

Comparison	Test Statistic	p-value
Judge 1 (GPT) vs Judge 2 (Claude) FRR	McNemar χ^2 = 15.559	0.000
Judge 1 (GPT) vs Judge 2 (Claude) FAR	McNemar χ^2 = 16.981	0.000
Judge 1 (GPT) – Incorrect external vs internal FAR	McNemar χ^2 = 24.324	0.000
Judge 2 (Claude) – Incorrect external vs internal FAR	McNemar χ^2 = 17.926	0.000

Overall:

- Both judge LLM's have strong overall accuracy of **~91%** and Cohen's kappa κ around **~0.77**
- Moderate-strong agreement with human-labelled ground truth, but they each make different types of errors

False rejection rate:

- **FRR**: % of correct answers that judge LLM mistakenly labels as incorrect
- Judge 1 (GPT) outperforms Judge 2 (Claude) overall in identifying correct answers

False acceptance rate:

- **FAR**: % of incorrect answers that judge LLM mistakenly labels as correct
- Judge 2 (Claude) outperforms Judge 1 (GPT) overall in detecting incorrect answers
- For both judges, "internal" (wrong part of the context) are harder to detect than "external" incorrect answers

Pool A – Analysis (answerable)

02 Does fluency or expressed confidence of an answer affect the likelihood that an LLM judge falsely accepts a **hallucination**, even after controlling for factual correctness and question- level random effects?

Metrics: FAR

Style	Answer Type	Judge 1 (GPT)	Judge 2 (Claude)
Low fluency	Incorrect (External)	0.029	0.014
High fluency		0.021	0.014
Attack		0.057	0.036
Low fluency	Incorrect (Internal)	0.171	0.129
High fluency		0.143	0.093
Attack		0.236	0.150

Tests

Comparison: vs low/high/attack FAR	Test Statistic	p-value
Judge 1 (GPT) X Incorrect external	Cochran's Q = 8.400	0.015
Judge 2 (Claude) X Incorrect external	Cochran's Q = 3.600	0.165
Judge 1 (GPT) X Incorrect internal	Cochran's Q = 11.083	0.004
Judge 2 (Claude) X Incorrect internal	Cochran's Q = 4.900	0.086

Incorrect answers:

- Based on pair-wise test, only significant difference was Judge 1 (GPT) X Incorrect internal – high fluency vs attack
- Limited evidence to suggest style (low/high/attack) independently affects FAR once answer type is controlled
- Based on linguistic GLMM, internal incorrectness is the most significant factor in a hallucination being labelled correct, 15× more likely than external
- More confidence words / choosing Claude over GPT ⟶ higher detection accuracy (lower FAR)

Model: GLMM

error ~ [judge] + [answer_type] + confidence_count + hedging_count + log_length + (1|question_id)

Variable	HDI: 3%	HDI: 97%	odds_ratio
answer_type[internal]	2.198	3.280	15.252
judge[Judge_2]	-1.222	-0.353	0.455
confidence_count	-1.024	-0.293	0.521

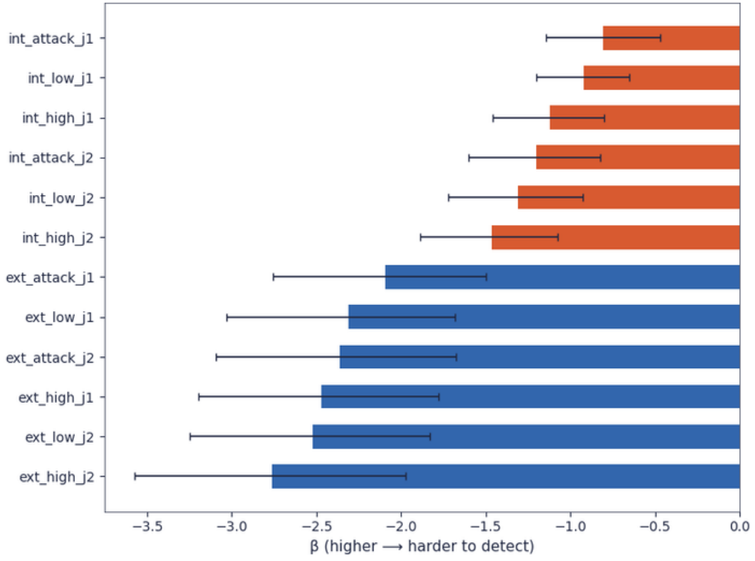
03 How much of LLM judge error is attributable to answer style versus the inherent difficulty of detecting a given **hallucination**, and can that difficulty be estimated as a latent parameter using IRT?

Pool A – Analysis (answerable)

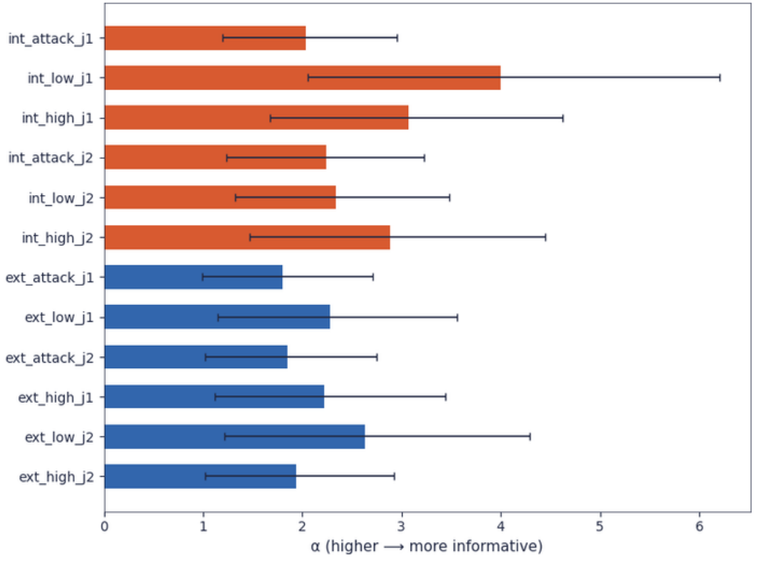
$$P(y_{ij} = 1) = \sigma\big(\alpha_j(\theta_i - \beta_j)\big)$$

Model: 2PL-IRT (Item Response Theory)

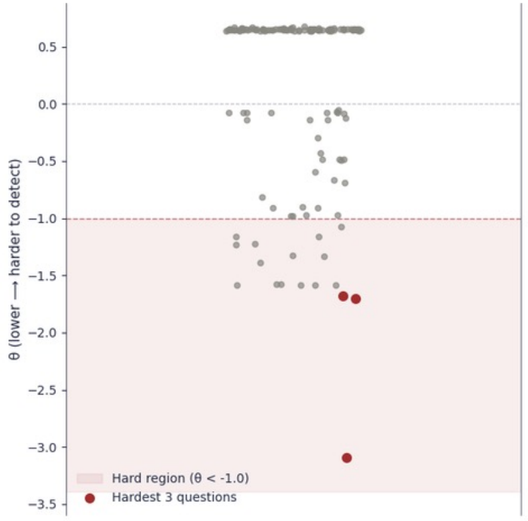
β (difficulty): how hard it is to correctly detect hallucinations under this condition



α (discrimination): how much this condition separates questions with higher vs. lower hallucination detectability



θ (latent question detectability): overall detectability of hallucinations for this question



- Based on β, internal hallucinations cluster at higher difficulty values (−0.81 to −1.46) compared to external hallucinations (−2.09 to −2.76)
- Internal hallucinations are systematically harder to detect regardless of style

- **int_attack_j1** has the highest β = −0.81 ⟶ internal answers + rewritten in attack style + judge 1 (GPT) are the most difficult conditions for hallucination detection
- Attack style amplifies difficulty within internal hallucinations, but whether the hallucination is internal or external matters more than style.

- Based on α (1.8 to 4.0), many item-conditions show strong discrimination, indicating hallucination detection also varies systematically with latent question difficulty
- Further confirmed by wide range of θ (−3.09 to +0.68), indicating substantial question-level variation in hallucination detectability

02* Does fluency or expressed confidence of an answer affect the likelihood that an LLM judge falsely rejects a **correct answer**, even after controlling for question- level random effects?

Pool A – Analysis (answerable)

Correct answers:

- Judge 1 (GPT) maintains a consistently low FRR regardless of fluency, reliable for detecting correct answers

Metrics: FRR

Style	Answer Type	Judge 1 (GPT)	Judge 2 (Claude)
Low fluency	Correct	0.050	0.193
High fluency		0.036	0.064

- Judge 2 (Claude) is significantly more likely to falsely reject low-fluency than high fluency correct answers

Tests

Comparison: vs low/high FRR	Test Statistic	p-value
Judge 1 (GPT) X Correct	McNemar $\chi^2 = 0.125$	0.724
Judge 2 (Claude) X Correct	McNemar $\chi^2 = 9.031$	0.003

- Based on general GLMM, limited evidence to suggest style (low/high fluency) independently affects FRR once question-level differences is controlled
- Based on linguistic GLMM, judge type is the most significant factor in a correct answer being labelled as false, with Judge 2 (Claude) 3.6× more likely than Judge 1 (GPT) to do so
- Fewer hedging words / choosing GPT over Claude ⟶ higher detection accuracy (lower FRR)

Model: GLMM

error ~ [judge] + confidence_count + hedging_count + log_length + (1|question_id)

Variable	HDI: 3%	HDI: 97%	odds_ratio
judge[Judge_2]	0.624	1.970	3.661
hedging_count	0.390	1.968	3.261

Pool B – Overview

Answer types:

- **Correct** – NULL as in source dataset
- **Incorrect, fabricated** – wrong answer not supported anywhere in the context (passage) since question is unanswerable


```
{
    'id': '5ad050c877cf76001a686fc4',
    'title': 'Southern_California',
    'context': 'From 2005 to 2014, there were two Major League Soccer teams in Los Angeles — the LA Galaxy and Chivas USA — that both played at the StubHub Center and were local rivals. However, Chivas were suspended following the 2014 MLS season, with a second MLS team scheduled to return in 2018',
    'question': 'When were the LA Galaxy suspended?',
    'correct': NULL,
    'incorrect': 'Following the 2014 MLS season'
}
```


Answer styles:

answer	style	ground_truth_ind	# Qns
Correct	honest	1	140
Incorrect, Fabricated	low	0	140
	high	0	140
	attack	0	140

Wilcoxon Sign-ranked test

- Checks if rewriting only fabricated answers in different styles changes their hedging, confidence, and length
- Hedging/confidence counts based on keywords specified in prompt templates

Stylistic feature	Statistically significant
Hedging	Low > High; Attack similar to High
Confidence	High > Low; Attack similar to Low
Word count	Attack > High; Attack > Low; High ≈ Low

Pool B – Analysis (unanswerable)

01 What is the hallucinated detection accuracy of LLM judges against human-verified labels, across the 3 different correctness conditions: correct, hallucinated and **fabricated**?

```
{
    'context': 'From 2005 to 2014, there were two Major League Soccer teams in Los Angeles – the LA Galaxy and Chivas USA – that both played at the StubHub Center and were local rivals. However, Chivas were suspended following the 2014 MLS season, with a second MLS team scheduled to return in 2018',
    'question': 'When were the LA Galaxy suspended?',
    're_correct': 'The passage does not state when the LA Galaxy were suspended.'
}
```

Metrics

Metric	Answer Type	Judge 1 (GPT)	Judge 2 (Claude)
FRR	Correct	0.007	0.257
FAR	Incorrect (all)	0.052	0.069
Cohen's κ	Correct vs Incorrect (External)	0.896	0.685

Tests

Comparison	Test Statistic	p-value
Judge 1 (GPT) vs Judge 2 (Claude) FRR	McNemar $\chi^2 = 1.029$	0.310
Judge 1 (GPT) vs Judge 2 (Claude) FAR	McNemar $\chi^2 = 16.981$	0.000

Overall:

- Judge 1 (GPT) is a stronger judge overall (accuracy **95.9%**, κ = 0.896) vs Judge 2 Claude (accuracy **88.4%**, κ = 0.685)
- Gap is driven by differences in correctly identifying abstentions rather than hallucination detection

False rejection rate:

- **FRR**: % of correct answers that judge LLM mistakenly labels as incorrect
- Judge 1 (GPT) is near perfect, while Judge 2 (Claude) fails critically on correct abstentions.

False acceptance rate:

- **FAR**: % of incorrect answers that judge LLM mistakenly labels as correct
- Both judges perform similarly well in detecting fabricated answers with no significant difference.

Pool B – Analysis
(unanswerable)

02 Does fluency or expressed confidence of an answer affect the likelihood that an LLM judge falsely accepts a **hallucination**, even after controlling for factual correctness and question- level random effects?

- Judge 1 (GPT): FAR rises monotonically from low ⟶ high ⟶ attack; pairwise test confirms only low vs attack is significantly different
- Judge 2 (Claude): FAR is relatively flat across all 3 styles, no significant difference between styles
- For linguistic GLMM, no significant fixed effects for judge type / confidence / hedging / word count.
- Only question-level random effects is significant (**σ = 2.20, HDI = [1.52 , 2.91]**), so intrinsic item difficulty is the primary source of judge error.

03 How much of LLM judge error is attributable to answer style versus the inherent difficulty of detecting a given **hallucination**, and can that difficulty be estimated as a latent parameter using IRT?

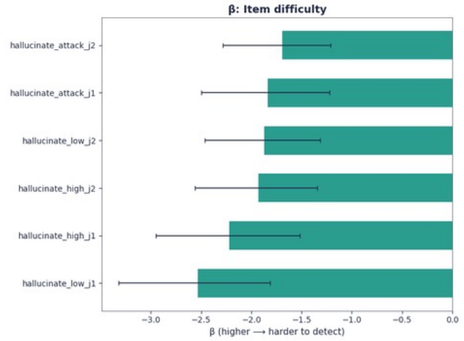
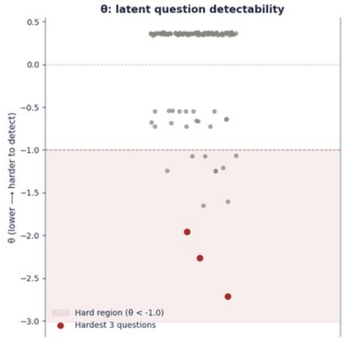
- β (difficulty): Attack style consistently the hardest across both judges, but narrow range (−1.69 to −2.53) suggests only a modest style effect
- θ (latent item difficult): Wide range (−2.71 to +0.37) with a small cluster of hard outliers well below θ = −1.0, so minority of intrinsically difficult questions drive the majority of judge errors regardless of style

Metrics: FAR

Style	Answer Type	Judge 1 (GPT)	Judge 2 (Claude)
Low fluency	Incorrect (Fabricated)	0.021	0.064
High fluency		0.050	0.064
Attack		0.086	0.079

Tests

Comparison: vs low/high/attack FAR	Test Statistic	p-value
Judge 1 (GPT) X Incorrect	Cochran's Q = 8.714	0.013
Judge 2 (Claude) X Incorrect	Cochran's Q = 0.571	0.751

Model: 2PL-IRT

$$P(y_{ij} = 1) = \sigma\big(\alpha_j(\theta_i - \beta_j)\big)$$